

Residency Endpoints for Logical Matrix–Vector Tasks

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Reference. A logical weight matrix $W \in \mathbb{Z}^{N \times K}$ is partitioned over resident 128×128 service units. An input slice is counted once at each receiving tile, so one vector evaluation receives

$$Q_S = \left\lceil \frac{N}{128} \right\rceil K \text{ B.}$$

Each valid resident weight element is written once when loading is included: $Q_R = NK$ B for per-use reloaded, and $Q_R = 0$ for weight-static.

Table II(a): Resident–streaming demand at two residency endpoints.

Residency	128×128	1024×1024	4096×4096	1024×4096	4096×1024
Weight-static	$Q_S = 128 \text{ B}$	$Q_S = 8 \text{ KiB}$	$Q_S = 128 \text{ KiB}$	$Q_S = 32 \text{ KiB}$	$Q_S = 32 \text{ KiB}$
	$Q_R = 0 \text{ B}$	$Q_R = 0 \text{ B}$	$Q_R = 0 \text{ B}$	$Q_R = 0 \text{ B}$	$Q_R = 0 \text{ B}$
	$\text{RI} = \infty$	$\text{RI} = \infty$	$\text{RI} = \infty$	$\text{RI} = \infty$	$\text{RI} = \infty$
Per-use reloaded	$Q_S = 128 \text{ B}$	$Q_S = 8 \text{ KiB}$	$Q_S = 128 \text{ KiB}$	$Q_S = 32 \text{ KiB}$	$Q_S = 32 \text{ KiB}$
	$Q_R = 16 \text{ KiB}$	$Q_R = 1 \text{ MiB}$	$Q_R = 16 \text{ MiB}$	$Q_R = 4 \text{ MiB}$	$Q_R = 4 \text{ MiB}$
	$\text{RI} = 1/128$	$\text{RI} = 1/128$	$\text{RI} = 1/128$	$\text{RI} = 1/128$	$\text{RI} = 1/128$

The primary boundary sums valid logical payload at all 128×128 tile input receivers. Each column specifies the logical weight shape $N \times K$ (outputs $\times$ inputs), not a new physical macro. Each window evaluates one input vector; input and resident elements are one byte (INT8 reference). Weight-static assumes preloading before the window; per-use reloaded writes the complete weight matrix once within it. $\text{RI} = Q_S/Q_R$, with $\text{RI} = \infty$ for $Q_R = 0$. Here, all dynamic ratios equal $1/128 = 0.0078125$. 1 KiB = 1024 B; 1 MiB = 1048576 B.

Interpretation. All five shapes have output dimensions divisible by 128. Their local-receiver ratios therefore coincide at $1/128$ despite different demand volumes and tile counts. The two rectangular tasks also have the same local-receiver Q_S and Q_R in this reference.

The contrast boundary, documented in the Chinese method and machine data, counts the whole-matrix input only once: $Q_S^{\text{op}} = K$ B and $\text{RI}^{\text{op}} = 1/N$ for per-use reloaded. It is not mixed into the table cells above.

Scope. One use is one vector evaluation. Per-use reloaded is neither an entire training step nor one load amortized over a multi-vector GEMM. Summed receiver demand is not a runtime prediction obtained by dividing by a single macro’s throughput. A future hardware comparison must match the service-unit set, write mode and precision.